

ANDROID APPLICATION DEVELOPMENT USING KOTLIN

- 1 University-level complete syllabus
- 2 Topic-wise structure from basic to advanced
- 3 Designed for exams, academics, and reference

1. Android Fundamentals

- 1 Introduction to Android
 - 2 History and evolution of Android
 - 3 Features of Android
 - 4 Android architecture
 - 5 Android versions and API levels
 - 6 Android ecosystem
 - 7 Types of Android applications
- a Linux Kernel
 - b Hardware Abstraction Layer (HAL)
 - c Android Runtime (ART)
 - d Native Libraries
 - e Application Framework
 - f Applications layer

2. Kotlin Programming for Android

- 1 Introduction to Kotlin
 - 2 Advantages of Kotlin
 - 3 Kotlin vs Java
 - 4 Structure of Kotlin program
 - 5 Variables and constants
 - 6 Data types and type inference
 - 7 Null safety
 - 8 Operators
 - 9 Control statements
 - 10 Functions
 - 11 Lambda expressions
 - 12 Collections
 - 13 Object-oriented concepts
- a Classes and objects
 - b Constructors
 - c Inheritance
 - d Interfaces
 - e Data classes
 - f Sealed classes

3. Android Studio and Project Structure

- 1 Android Studio overview
- 2 SDK Manager
- 3 AVD Manager
- 4 Project directory structure
- 5 Gradle build system
- 6 Build variants and flavors

- 7 AndroidManifest.xml
- 8 Resource directories
- 9 ViewBinding
- 10 DataBinding

4. Android Application Components

- 1 Activities
- 2 Intents
- 3 Services
- 4 Broadcast Receivers
- 5 Content Providers
- 6 Application class
- 7 Context

- a Activity lifecycle
- b Explicit and implicit intents
- c Foreground, background, and bound services
- d System and custom broadcasts

5. User Interface Development

- 1 Layouts
- 2 Views
- 3 Event handling
- 4 Styles and themes
- 5 Material Design components
- 6 Menus and dialogs

- a LinearLayout
- b RelativeLayout
- c ConstraintLayout
- d TextView
- e EditText
- f Button
- g ImageView

6. Fragments and Navigation

- 1 Fragments
- 2 Fragment lifecycle
- 3 Fragment transactions
- 4 Fragment communication
- 5 Navigation Component
- 6 Navigation graph
- 7 Bottom navigation
- 8 Navigation drawer

7. Lists and Advanced UI Components

- 1 RecyclerView
- 2 Adapter
- 3 ViewHolder

- 4 LayoutManager
- 5 DiffUtil
- 6 ListAdapter
- 7 Custom views
- 8 Animations
- 9 MotionLayout

8. Data Storage

- 1 SharedPreferences
 - 2 Internal storage
 - 3 External storage
 - 4 SQLite database
 - 5 Room database
 - 6 Firebase Realtime Database
 - 7 Cloud Firestore
 - 8 External databases via REST APIs
- a Entity
 - b DAO
 - c Database class

9. Networking and Web Services

- 1 Internet permissions
- 2 RESTful services
- 3 HTTP methods
- 4 Retrofit
- 5 OkHttp
- 6 JSON parsing
- 7 API authentication
- 8 Error handling

10. Background Processing

- 1 Threads
- 2 Handlers
- 3 AsyncTask (legacy)
- 4 Kotlin Coroutines
- 5 WorkManager
- 6 JobScheduler
- 7 Foreground services
- 8 Alarms

11. Android Architecture Patterns

- 1 MVC
- 2 MVP
- 3 MVVM
- 4 ViewModel
- 5 LiveData
- 6 Flow

- 7 StateFlow
- 8 Clean Architecture

12. Dependency Injection

- 1 Dependency Injection concepts
- 2 Hilt
- 3 Koin
- 4 Injecting ViewModels
- 5 Scopes
- 6 Best practices

13. Jetpack Compose

- 1 Compose basics
- 2 Composable functions
- 3 State management
- 4 Layouts in Compose
- 5 Navigation in Compose
- 6 Animations in Compose

14. Security and Permissions

- 1 Android permission model
- 2 Runtime permissions
- 3 Secure data storage
- 4 Encryption
- 5 Biometric authentication

15. Testing

- 1 Unit testing
- 2 Instrumentation testing
- 3 UI testing
- 4 Espresso
- 5 Mocking
- 6 Test-driven development

16. Performance Optimization

- 1 Memory management
- 2 Leak detection
- 3 App startup optimization
- 4 Battery optimization
- 5 Network optimization

17. Firebase and Cloud Services

- 1 Firebase Authentication
- 2 Cloud Firestore

- 3 Realtime Database
- 4 Cloud Messaging
- 5 Analytics
- 6 Crashlytics

18. Advanced Android Topics

- 1 Multi-module projects
- 2 Dynamic feature modules
- 3 Kotlin Multiplatform
- 4 Android NDK basics
- 5 Jetpack libraries

19. Deployment and Maintenance

- 1 App signing
- 2 Build variants
- 3 ProGuard and R8
- 4 Google Play Store publishing
- 5 App updates and maintenance

20. Project Development

- 1 Requirement analysis
- 2 System design
- 3 Implementation
- 4 Testing
- 5 Documentation
- 6 Mini and major projects